

Roll No.

TCS-812

**B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

GRAPH DATABASE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe Graph Database with suitable examples.
(b) Discuss any *five* applications of Graph Databases.
(c) What are the major challenges associated with the Graph Database ?
2. (a) What are the key differences between RDBMS and Graph Database ?
(b) Discuss the tools and techniques used to create Graph Databases.
(c) What are the main advantages and disadvantages of using Graph Databases ?
3. (a) What is Graph Data Modelling ? Explain with an example.
(b) Design the Graph Data Model for any *one* use case.
(c) Discuss Graph Database Modeling guidelines.

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4. (a) Consider a RDBMS model of employee management system and convert it into a Graph Database Model.
(b) Discuss the building blocks of Graph Databases.
(c) Discuss the building blocks of Neo4j.
5. (a) Discuss the main features of the Neo4j tool.
(b) Write the Neo4j commands for CQL functions and CQL Admin.
(c) Write Neo4j CQL command syntax for read clauses, write clauses and general clauses (three commands in each clause).

Roll No.

TCS-822

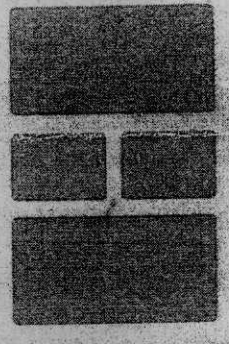
**B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

MOBILE APPLICATION DEVELOPMENT

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Observe the following GUI and write an XML file using relative layout to create the same : (CO1)

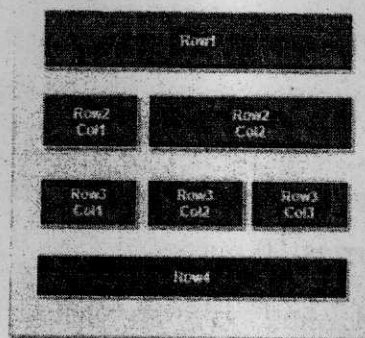


- (b) Describe the Android architecture in detail. (CO1)
(c) Discuss the need of permissions in Android. Describe the permissions to set system functionalities like Bluetooth, Camera. (CO1)

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2. (a) Explore the steps to install and configure Android Studio and SDK. (CO2)
- (b) Describe the significance of SQLite database in Android. (CO2)
- (c) Explain Date and Time picker with its methods. (CO2)
3. (a) Write the directory path where images are stored while developing Android applications. (CO3)
- (b) List sensors in Android and explain any *one* in detail. (CO3)
- (c) Write a program to display the following output using suitable layout : (CO3)



4. (a) When performing sanity test on the mobile application. What all criteria should be taken into consideration ? (CO4)
- (b) Define Testing. Explain Manual Testing and Automated Testing. (CO4)
- (c) Which things to consider testing a mobile application through black box technique ? (CO4)
5. (a) Write a program to display a circular progress bar. (CO5)
- (b) Write a program to locate user's current location. (Write ONLY .java and manifest file). (CO5)
- (c) Develop an application to send and receive SMS. (Write ONLY .java and permission). (CO5)

Roll No.

TCS-828

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
ADVANCE MICROPROCESSOR SYSTEM AND DESIGN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Draw and explain block diagram of 8255. (CO6)
- (b) Initialize 8255 in I/O mode. To take input from port C and finds its 2's complement and give output to port A. (CO6)
- (c) Interface ADC 0808 with 8086 using 8255 ports. Use port A of 8255 for transferring digital data output of ADC to CPU and port C for control signals.
Assume an analog input present at I/P2 of ADC. (CO6)
2. (a) Interface 32 KB of RAM memory to 8086 microprocessor system with suitable addresses. (CO6, CO5)

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- (b) What is the difference between RISC and CISC processor ? (CO6, CO5)
- (c) Differentiate between 80386 and 80486 processor. (CO6, CO5)
3. (a) Explain different interconnection topologies of Multiprocessors. (CO3, CO4)
- (b) What are the different addressing modes in 8086 ? Explain with example. (CO3, CO4)
- (c) Explain the meaning of the following 8086 instructions :
- (i) SUB [2345], BX
 - (ii) AND AX, [BX]
 - (iii) MOV BX, 2345
 - (iv) XOR AX, BX
4. (a) Draw block diagram of Intel 8086 microprocessor. (CO3)
- (b) Write an 8086 assembly language program to add two 16 bit numbers. (CO3)
- (c) Explain program development tools editor, assembler and linker. (CO3)
5. (a) What are the differences between 8086 and 8088 ? Explain memory organization in 8086. (CO2, CO1)
- (b) Explain numeric coprocessor 8087. (CO2, CO1)
- (c) Explain concept of virtual memory. What is paging and segmentation ? (CO2, CO1)

Roll No.

TCS-851

**B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

STORAGE NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each question carries 10 marks.

1. (a) What is unstructured data ? Illustrate the core elements that are essential for the basic functionality of a data center. (CO1)
- (b) (i) How the value of data might change over time ? Explain with suitable example.
- (ii) Write a short note on IDE/ATA and SCSI. (CO1)
- (c) Explain the evolution of storage architecture with a neat diagram. (CO1)
2. (a) Explain FC-SAN. Discuss the advantage of FC-SW over FC-AL. (CO2)
- (b) What is DAS ? Illustrate DAS, benefits and limitations. (CO2)
- (c) What is virtualization ? Discuss the advantages and disadvantages of in-band and out-band implementation. (CO2)

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3. (a) A large company is considering a storage infrastructure-one that is scalable and provides high availability. More importantly, the company also needs performance for its mission-critical applications. Which storage topology would you recommend (SAN, NAS) and why ?
(CO3, CO6)
- (b) (i) Why do formatted disks have less capacity than unformatted disks ?
(ii) Which components constitute the disk service time ? Explain.
(CO3, CO6)
- (c) Describe RAID ? What are the benefits of using RAID3 in a backup application ?
(CO3, CO6)
4. (a) Illustrate block level storage virtualization with a neat diagram.
(CO4, CO6)
- (b) SAN is configured for a backup to disk environment and the storage configuration has additional capacity available. Can you have a NAS gateway configuration using this SAN ? Discuss the implications of sharing the backup-to-disk SAN environment with NAS. (CO4, CO6)
- (c) Illustrate different protocols available for transmitting storage data traffic over TCP/IP. Explain the use of these protocols. (CO4, CO6)
5. (a) Discuss network attached storage and IPSAN technologies in detail.
(CO5)
- (b) Explain how the performance of NAS can be affected if the TCP window size at the sender and the receiver are not synchronized. (CO5)
- (c) Illustrate HOT swap, Hot spare and RAID controller in implementing RAID.
(CO5)

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TCS-855

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023

AGILE SOFTWARE ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Enlist at least *four* agile methods. Why the agile methods are often referred to as incremental and evolutionary ? Discuss ideology of agile methodology. (CO1/CO2)
- (b) Briefly explain the central principles of Extreme Programming (XP). In the context of XP, explain the following terms : (CO1/CO2)
Refactoring and Continuous Integration.
- (c) Give an example of a project for which the agile model would be suitable and one project for which the agile model would not be appropriate. (CO1/CO2)

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2. (a) What is Scrum ? What goes inside Scrum Framework ? Who is responsible to manage it ? Explain using schematic diagram.(CO2/CO3)
- (b) Who are in the Scrum team and what are their roles ? How does the daily Scrum work ? (CO2/CO3)
- (c) Give an example of user story related to online shopping by a customer. Write the acceptance criteria for it. (CO2/CO3)
3. (a) Briefly state the symptoms of poor design of software. What are various object-oriented design principles that help developers eliminate the symptoms of poor design ? Explain briefly any *one* of such design principles. (CO4)
- (b) What is the Single Responsibility Principle (SRP) ? Explain with an example, how this principle helps software design to handle changes in requirements. (CO4)
- (c) What are the *two* primary attributes of the modules that conform to Open/Closed Principle (OCP) ? Although these *two* attributes are at odds, explain through a suitable example how a design can conform to the OCP. (CO4)
4. (a) Briefly explain the difference between unit tests and acceptance tests. Explain Test-Driven Development with an example. (CO5)
- (b) What are these terminologies in context to Agile Scrum : Sprint, Sprint Backlog, User story, Sprint plan, Daily stand-up meeting and Sprint review ? (CO5)

(3)

- (c) What is the concept of continuous integration and continuous deployment in Agile Scrum ? How is Agile lifecycle different from traditional lifecycle of Waterfall model as far as Quality Assurance (QA) testing is concerned ? (CO5)
5. (a) What are the challenges involved in Agile as far as customer relations are concerned in the market ? Briefly explain how these challenges are handled to make customer relations robust. (CO6)
- (b) Even if Scrum practices are really encouraging, why does then Scrum fail ? What are the gains even if it fails ? (CO6)
- (c) Explain briefly regarding Agile projects on Cloud. Throw some light on Agile rapid development technologies. (CO6)

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TCS-812

**B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

GRAPH DATABASE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe Graph Database with suitable examples.
(b) Discuss any *five* applications of Graph Databases.
(c) What are the major challenges associated with the Graph Database ?
2. (a) What are the key differences between RDBMS and Graph Database ?
(b) Discuss the tools and techniques used to create Graph Databases.
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(b) Design the Graph Data Model for any *one* use case.
(c) Discuss Graph Database Modeling guidelines.

P. T. O.

(2)

4. (a) Consider a RDBMS model of employee management system and convert it into a Graph Database Model.
(b) Discuss the building blocks of Graph Databases.
(c) Discuss the building blocks of Neo4j.
5. (a) Discuss the main features of the Neo4j tool.
(b) Write the Neo4j commands for CQL functions and CQL Admin.
(c) Write Neo4j CQL command syntax for read clauses, write clauses and general clauses (three commands in each clause).

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TCS-859

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
SERVICE ORIENTED CLOUD ARCHITECTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Bosch Thermotechnology North America developed its first cloud connected device using AWS Lambda and AWS IoT Core, optimizing costs while improving customer experience. Evaluate the types of Magento module dependencies with suitable case study of above mentioned organization. (CO1)
- (b) Education Technology Company Vocareum provided its customers with the option of using Amazon Lightsail to spin up new test environment quickly and cost efficiently for students to practice cloud skills. Evaluate web services framework with suitable case study of above mentioned organization.

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TCS-859

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
SERVICE ORIENTED CLOUD ARCHITECTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Bosch Thermotechnology North America developed its first cloud-connected device using AWS Lambda and AWS IoT Core, optimizing costs while improving customer experience. Evaluate the types of Magento module dependencies with suitable case study of above mentioned organization. (CO1)

(b) Education Technology Company Vocareum provided its customers with the option of using Amazon Lightsail to spin up new test environments quickly and cost efficiently for students to practice cloud skills. Evaluate web services framework with suitable case study of above mentioned organization.

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- (c) Educational Technology Company Sumadi delivers near-real-time exam supervision simply and cost effectively after migrating to AWS and was able to scale to support 50,000 concurrent users. Evaluate technology related to full stack development with suitable case study of above mentioned organization. (CO1)
2. (a) Ascend Money created a multi-arch approach to cloud computing using Amazon EC2 Instances, improving performance upto 40 percent while lowering costs and freeing developers to innovate. Evaluate Components of SOA with suitable example of above mentioned organization. (CO2)
- (b) Dynata uses Amazon Pinpoint to centrally manage communications, automate campaign management, and drive better customer engagement through improved visibility. Evaluate advantages and disadvantages of SOA with suitable case study of above mentioned organization. (CO2)
- (c) Thomson Reuters,. a global content-driven technology company, reduced recovery point objective time from 2 hours to 30 minutes by migrating its single sign-on authentication solution to AWS. Evaluate enterprise service bus with suitable case study of above mentioned organization. (CO2)
3. (a) Comprinno Technologies, a professional services startup, uses the AWS Well-Architected Framework to standardize the sales process, provide direction for customers, and grow its business. Evaluate SOA vs. microservices with suitable case study of above mentioned organization. (CO3)

- (b) CarbonCure uses AWS to turn data into insights, so that it can help customers reduce embodied carbon from concrete in new construction. Evaluate components of Web Service with suitable case study of above mentioned organization. (CO3)
- (c) NCS migrated its on-premises service desk solution to Amazon connect to halve onboarding time, reduce operations costs, and improve customer communications with new technologies such as artificial intelligence and machine learning. Evaluate characteristics of Web Service with suitable case study of above mentioned organization. (CO3)
4. (a) Work Management Company Smartsheet migrated its on-premises data centers to Amazon RDS and increased reliability, scalability, flexibility, and agility, improving performance for customers globally. Evaluate SLA Performance metrics with suitable case study of above mentioned organization. (CO4)
- (b) Innovyze, an Autodesk company, reduced development time from years to months and deployment time from months to days by building Info360, which uses AWS IoT Greengrass and runs entirely on the cloud. Evaluate the multilevel SLA structure with suitable case study of above mentioned organization. (CO4)
- (c) Rackspace Technology, an AWS Partner, used AWS Training and Certification to build training programs to certify employees and improve sustainability, growth, and staff productivity. Evaluate accessibility of the business cloud techniques with suitable case study of above mentioned organization. (CO4)

5. (a) Leumi Group used Amazon EKS and Amazon EKS anywhere to simplify the migration of its on-premises solutions to the cloud, saving time and money, supporting innovation, and accelerating time to market. Evaluate consumer device ecosystems and cloud features of Microsoft, Apple, Google and Dropbox with suitable case study of above mentioned organization. (CO5)
- (b) Sports Gaming Company FanDuel migrated four live stream linear channels to AWS in less than 3 weeks, increasing its reliability, performance, and security for a better customer experience. Evaluate business and consumer cloud benefits and risks for people with disabilities with suitable case study of above mentioned organization. (CO5)
- (c) Water Technology Company Xylem has been able to reduce employee onboarding time and achieve a comprehensive view of security across its AWS accounts by using AWS IAM Identity Center. Evaluate transfer security modes of WCF service with suitable case study of above mentioned organization. (CO5)

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TDM-881

**B. TECH. (CSE/IT) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the terms Disaster, Hazard and Risk. Also discuss the role of governance in mitigating the effects of disasters. (CO1, CO3)
- (b) What is Vulnerability ? Discuss the relation between the vulnerability and risk. (CO1, CO3)
- (c) Discuss the various epidemics caused by the natural disasters. Differentiate Natural and Man-made disasters. (CO1, CO3)
2. (a) What are the causes of Earthquakes ? Discuss the measures to mitigate the impact of this disaster. (CO2, CO4)
- (b) What are landslides ? List out major causes of landslides in India. (CO2, CO4)

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(2)

- (c) Why is drought a slow onset disaster ? Identify *five* factors that cause droughts. (CO2, CO4)
3. (a) What is risk ? Discuss the factors affecting vulnerabilities. State how the weaker groups in the society are more vulnerable. (CO3, CO5)
- (b) Write a note about the natural disasters. When do natural events become natural hazards ? (CO4, CO5)
- (c) When do cyclones occur in India ? Describe the measures adopted for protection from the cyclones. (CO3, CO5)
4. (a) What is Disaster Management ? Explain the various phases of Disaster Management Cycle. (CO3, CO4)
- (b) What is disaster preparedness ? Discuss how ICT can be helpful in disaster preparedness. (CO3, CO4)
- (c) What is disaster recovery ? Analyse the challenges associated with disaster recovery. (CO3, CO4)
5. (a) Explain the role of National Disaster Management Authority (NDMA), Local government, Society for the mitigating and restoring from the effects of disaster. (CO3, CO5)
- (b) Define Disaster according to Disaster Management Act, 2005. Explain basic format of the Disaster Management Cycle. (CO3, CO5)
- (c) Propose a suitable Disaster Information Management System (DIMS) to support the processes during the various phases of Disaster Management. (CO3, CO5)

Roll No.

TCS-881

**B. TECH. (CSE-SE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

SOFTWARE MAINTENANCE AND REUSE

Time : THREE Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are some of the common tools used for software maintenance and how do they assist in tasks such as bug fixing, code refactoring and version control ?
(CO1, CO2, CO3)
- (b) What are the key components of a software maintenance framework, and how do they contribute to effective software maintenance practices ?
(CO1, CO2, CO3)
- (c) Define the following :
(CO1, CO2, CO3)
 - (i) Information gap
 - (ii) Maintenance challenge
 - (iii) Operating environment
 - (iv) Maintenance personnel
 - (v) Software maintenance framework

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2. (a) What are the personnel aspects that can impact software maintenance activities and how can software development teams manage these factors ? (CO1, CO2, CO3)
- (b) Write a note on the significant factors of software maintenance process. (CO1, CO2, CO3)
- (c) How do new development activities differ from maintenance activities in software engineering and why is software maintenance important for ensuring the long-term success of software systems ? (CO1, CO2, CO3)
3. (a) How do the different factors that impact software maintenance interact with each other ? (CO1, CO2, CO3)
- (b) What are the different types of software changes that can occur during the maintenance phase ? (CO1, CO2, CO3)
- (c) What are Lehman's eight laws of software evolution and how do they describe the evolutionary process of software systems overtime ? (CO1, CO2, CO3)
4. (a) What are the key objectives and benefits of using reverse engineering in software development ? (CO4, CO5)
- (b) What are the main objectives and benefits of reverse engineering in various industries and how can this process be used to improve product design ? Identify intellectual property infringement and enhance product maintenance and repair. (CO4, CO5)
- (c) What are the different levels of reverse engineering and how does each level differ in terms of complexity, time and resources required ? (CO4, CO5)

(3)

5. (a) What are the different types of software restructuring and how can they be used to improve the quality, maintainability and scalability of software systems ? (CO4, CO5)
- (b) What are the main inputs and outputs involved in the domain analysis ? (CO4, CO5)
- (c) Describe the characteristics of Reusable Component. (CO4, CO5)

Roll No.

TCS-891

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the different challenges towards computer security ? (CO1)
- (b) What are the different security mechanisms ? Explain any *two* of them.
(CO1)
- (c) What do you mean by the term Cryptography ? How is it different from cryptanalysis ?
(CO1)
2. (a) What is the purpose of diffusion in the design of block ciphers ? (CO2)

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TCS-891

B. TECH. (CSE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the different challenges towards computer security ? (CO1)
- (b) What are the different security mechanisms ? Explain any *two* of them.
(CO1)
- (c) What do you mean by the term Cryptography ? How is it different from cryptanalysis ? (CO1)
2. (a) What is the purpose of diffusion in the design of block ciphers ? (CO2)

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(2)

- (b) Define any *two* of the following terms : (CO2)
- (i) Steganography
 - (ii) Brute force attack
 - (iii) Gatekeeper function
- (c) Explain any *two* modes of operations of Block Ciphers. (CO2)
3. (a) Describe the various methods of distributing session keys. What is the role of Key Distribution Centre ? Explain with neat diagram. (CO3)
- (b) Write an explanatory note on RSA. (CO3)
- (c) Describe the triple DES with *three* DES keys. (CO3)
4. (a) What are the *three* approaches to message authentication ? Discuss each of them briefly. (CO4)
- (b) Write an explanatory note on X.509 directory authentication service. (CO4)
- (c) What are the two protocols used in IP Security ? Explain each of them. (CO4)
5. (a) Explain some of the well-known Cloud Security risks and Counter-measures. (CO5)
- (b) Write some of the characteristics of the firewall. (CO5)
- (c) What are the Packet Filtering Firewalls ? What are the different filtering rules for them ? (CO5)

Roll No.

TIT-801

**B. TECH. (IT) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

IT INFRASTRUCTURE MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the various elements that make up IT infrastructure. Briefly explain the development of Internet. (CO1)
- (b) What are the common tasks of IT System Management ? (CO1)
- (c) Define the term 'Total Cost of Ownership' (TCO) and 'Return on Investment' (ROI). What are the significance of these terms in context of IT service management ? (CO1)
2. (a) What are the important steps involved in implementation of service level management ? (CO1, CO2)
- (b) What is financial management ? What are the various activities of financial management process ? (CO1, CO2)

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- (c) What is cost-benefit analysis ? Discuss the following terms used in cost-benefit analysis : (CO1, CO2)
Payback period, Net present value, Return on investment.
3. (a) Discuss briefly the *four* important processes involved in the implementation of IT service continuity management. (CO3, CO4)
(b) What is IT service continuity management ? What are the advantages it offers to an organization implementing it ? (CO3, CO4)
(c) What are the main problem that occur while implementing configuration management function ? (CO3, CO4)
4. (a) What is difference between incident management and problem management ? (CO4, CO5)
(b) Why is release management important ? What are the benefits it offers to an IT organization in providing services to its customers ? (CO4, CO5)
(c) What are the need of storage management ? Explain the process activities of storage management. (CO4, CO5)
5. (a) Explain the activities involved in DRP (Disaster Recovery Plan). Give *five* different methods of testing the DRP. (CO5, CO6)
(b) What is the difference between identification and verification biometric systems ? (CO5, CO6)
(c) Mention various types of privacy laws and discuss information privacy law in detail. (CO5, CO6)

Roll No.

TCS-802

**B. TECH. (CS/IT) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

MOBILE COMPUTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe WLAN-based architecture. How does a WLAN function ?
(CO1)
- (b) What are the parameters considered for defining QoS in GPRS technology ?
(CO1)
- (c) Specify advantages of static channel allocation over dynamic channel location strategies.
(CO1)
2. (a) In context to cellular network, discuss the following : (CO1, CO2)
Cluster, Re-use distance, Cell splitting and Sectorisation

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- (b) What is a MSC ? What are the functions of MSC in network and switching subsystem ? (CO1, CO2)
- (c) What are the problems faced when using Transmission Control Protocol for Mobile Adhoc Networks ? (CO1, CO2)
3. (a) What are issues in data management ? Describe data replication for mobile computer. (CO3)
- (b) What do you mean by multihopping ? Explain multicluster architecture and problems related to this architecture. (CO3)
- (c) Why is coda promising and potentially very important ? Discuss about RPC2 and side effect in coda file system. (CO3)
4. (a) Discuss life cycle of mobile agent-based computing with suitable diagram. (CO4)
- (b) What is security and fault tolerance issues in the context of Mobile Agents ? (CO4)
- (c) Discuss some suitable applications of mobile agent along with their advantages and disadvantages. (CO4)
5. (a) Why is it not possible to use circuit switching in Adhoc Networks ? What are the advantages and disadvantages of proactive protocols ? (CO5, CO6)
- (b) Explain about Ad-hoc On-demand Distance Vector routing protocols. Why is it more prominent in adhoc network scenario ? (CO5, CO6)
- (c) Explain the caching policies for DSR Protocol along with benefits and drawback. (CO5, CO6)